Circulation

ORIGINAL RESEARCH ARTICLE

Accelerometer-Measured Physical Activity After Mitral Valve Surgery: An Analysis of the UK Mini Mitral Randomized Controlled Trial

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BACKGROUND: Wearable accelerometer devices measure free-living physical activity and sleep without relying on self-reports. Their utility to measure and compare recovery of physical function after cardiac surgery procedures has not been previously studied in the setting of a randomized controlled trial.

METHODS: Data were collected during the UK (United Kingdom) Mini Mitral trial, in which patients were randomized to undergo either a sternotomy or a minimally invasive thoracoscopically guided right minithoracotomy procedure (Mini) for mitral valve repair. This is a secondary analysis of the trial data using a different primary end point. Patients wore a wrist-worn triaxial accelerometer on their nondominant wrist for 24 hours over a 7-day period before surgery and at 6, 12, 18, 24, 38, and 52 weeks after surgery. Accelerometer outcomes included the change from baseline to 52 weeks in total activity counts and time spent in moderate-to-vigorous physical activity, light physical activity, and sedentary time. Time spent asleep and sleep efficiency were also captured. Accelerometry data were processed and analyzed by researchers blinded to the surgical approach.

RESULTS: A total of 230 patients (115 in each trial arm) provided valid accelerometry data. There were significant differences between arms in total activity counts; the mean difference was 26 744 (95% CI, 9085–44 403; P=0.003) at 6 weeks and 26 060 (95% CI, 6971–45 149; P=0.008) at 18 weeks, both favoring Mini. Time spent in moderate-to-vigorous physical activity also favored the Mini arm at 6 and 18 weeks, with mean differences of 15 (95% CI, 5.7–24; P=0.001) and 9.9 (95% CI, 0.31–20; P=0.043) minutes per day, respectively. Those in the Mini arm also had significantly lower sedentary time (at 12, 18, and 24 weeks) and spent more time in light physical activity (at 18 weeks) than those in the sternotomy arm. There were no differences in sleep duration between arms at any time point, although those in the Mini arm had higher sleep efficiency than those in the sternotomy arm at 12 weeks.

CONCLUSIONS: This analysis from the UK Mini Mitral trial suggests that wearable accelerometer devices can be used to compare recovery after surgical procedures. The data support an overall decline from baseline activity to 52 weeks in patients undergoing a sternotomy and an earlier recovery of physical activity after a minimally invasive thoracoscopically guided right minithoracotomy approach.

Key Words: accelerometry ■ thoracic surgery

The rate of recovery of physical function after surgery is a key factor for patients when making decisions about choosing different surgical approaches.[1] Minimally invasive approaches are increasingly used in all surgical specialties as clinicians and patients seeks to reduce the surgical insult and accelerate the time taken to recover physical function after surgery.

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Clinical Perspective

What Is New?

- The utility of wearable accelerometer devices to measure and compare recovery of physical function after surgical procedures, particularly in cardiac surgery, has not been previously studied.
- This analysis shows that wearable accelerometer devices can be used to chart recovery of physical function after surgery and to compare recovery after different surgical procedures.
- It demonstrated earlier recovery of physical activity and improved sleep efficiency after minimally invasive thoracotomy compared with sternotomy for mitral valve repair surgery.

What Are the Clinical Implications?

- Objective data from wearable devices are less prone to bias and measurement error compared with self-reported measures. Researchers should consider their use to provide additional or complementary information when designing trials to compare surgical procedures.
- When faced with decisions about surgery, patients are often worried about recovery time. This analysis shows that minimally invasive mitral valve repair by a minithoracotomy has a smaller and shorter impact on patients' physical activity and functioning.

Nonstandard Abbreviations and Acronyms

Mini	minimally invasive thoracoscopically guided right minithoracotomy procedure
MVPA	moderate-to-vigorous physical activity
SF-36v2	36-Item Short Form Health Survey version 2.0

The most common minimally invasive cardiac procedure is mitral valve repair surgery, usually performed by a minimally invasive thoracoscopically guided right minithoracotomy (Mini). The Mini procedure is performed through a 4- to 7-cm lateral thoracotomy, in contrast to conventional mitral valve repair surgery, which is performed by a full sternotomy. The full sternotomy incision is closed using multiple sternal wires, bands, or plates, immobilizing the sternum, and sternal union occurs at around 12 weeks after surgery.[2] During this period, patients are significantly limited in physical activity and daily functioning, which can increase the risk of complications.[3]

The Mini approach involves a smaller incision than conventional sternotomy, but widespread adoption of the technique has been limited by uncertainty about the safety and efficacy and lack of evidence of faster recovery of physical function.[4] The recently published UK Mini Mitral trial, the largest randomized controlled trial comparing Mini versus conventional sternotomy, established the safety and efficacy of the Mini technique for the first time in a large multicenter randomized controlled trial.[5] The trial also compared self-reported physical functioning and return to usual activities, measured using the 36-Item Short Form Health Survey version 2.0 (SF-36v2) physical functioning scale.[6] No significant difference in self-reported physical functioning was found between those who received Mini versus sternotomy up to 12 weeks after the index surgery.[5]

Current established measures of recovery of physical function after surgery are validated questionnaires such as the SF-36v2 or the European Organisation for Research and Treatment of Cancer Core Quality of Life questionnaire (EORTC QLQ-C30). However, for assessment of surgical outcomes in clinical trials, they are limited by the potential for bias caused by challenges in blinding patients after surgery, which cannot be easily overcome. Moreover, self-reported measures of physical activity are not well-suited to capturing changes in physical activity levels over time[7] and have high measurement error, limiting the interpretability of findings.[8]

Accelerometers capture free-living movement of all intensities, usually over a week-long period, and can be used to measure physical activity. There are multiple methodological considerations when using accelerometers in research contexts, including location on the body they will be worn (eg, wrist, waist, thigh, or ankle). Wrist-worn accelerometers are becoming increasingly popular in epidemiological and clinical studies because they can be worn 24 hours per day, enabling measurement of both waking activities and sleep and improving wear compliance by participants.[9] Wrist-worn accelerometers have good validity and reliability for the measurement of physical activity and sleep[10,11] and have higher measurement validity than subjective methods.[12] They may provide more reliable or complementary information when used to assess recovery after surgical procedures.

The aims of the study were: (1) to determine the utility of accelerometery to measure recovery of physical function and sleep after mitral valve repair surgery, and (2) to compare recovery of physical activity from baseline to 52 weeks in patients undergoing Mini versus conventional sternotomy for mitral valve repair as part of the UK Mini Mitral trial.

METHODS

UK Mini Mitral was a pragmatic, multicenter, randomized controlled trial comparing patients undergoing Mini versus conventional sternotomy for mitral valve repair. Expertise-based randomization with surgeons only performing the operation for which they were experts enabled the robust comparison of the 2 techniques. Full details of the trial protocol[13] and primary results of the trial[5] have been published. Ethical approval was provided by National Health Service Wales ethics committee 6 (16/WA/0156). Deidentified data are available from the study chief investigator (E.F.A.) upon reasonable request.

In a prespecified substudy, which started 1 year after the main trial opened to recruitment, randomized participants were asked to wear a GENEActiv triaxial accelerometer recording at a sampling rate of 100 Hz on their nondominant wrist nonstop for 7 consecutive days at 7 time points during the trial: baseline (presurgery) and 6 weeks, 12 weeks, 18 weeks, 24 weeks, 38 weeks, and 52 weeks after index surgery. Accelerometry data collection was paused at the onset of the COVID-19 pandemic, from March 7, 2020, to October 31, 2020. Accelerometry files were downloaded using the proprietary software and were exported into 60-s epoch.csv files.

A researcher trained in accelerometry who was blinded to participant allocation processed the accelerometry files. Following best practice, a trace of each raw accelerometer file was visually inspected before processing for quality assurance (ie, to note nonwear, device failure, spurious data, etc). The first and last days of wear were removed from the data set. Files were then processed using the Sedentary Sphere,[14,15] in which each minute was classified as 1 of 2 physical behaviors (sleep, moderate-to-vigorous physical activity [MVPA], light physical activity, or sedentary time) using a combination of intensity cut points (detailed below) and arm-angle inclination. To our knowledge, no automated algorithms have been validated for the identification of the sleep window among cardiac patients. The sleep window was therefore manually identified through visual inspection of raw accelerometer traces and heatmaps in line with best practice.[16] There were several accelerometry files (6.6%) in which the sleep period could not be clearly identified; in these cases, only MVPA was extracted from the file and retained in analysis. Sleep efficiency was determined based on the number of minutes within the sleep window classed as sleep, with possible values theoretically (but never practically) ranging from 0 (0 sleep efficiency) to 1 (perfect sleep efficiency).

Table 1. Description of Accelerometer Substudy Participants

Characteristic	Minithoracotomy (n=115)	Conventional sternotomy (n=115)
Demographic		
Age at randomization (y), mean (SD)	67.3 (9.9)	66.4 (11.8)
Sex, No. (%)		
Male	86 (74.8)	77 (67.0)
Female	29 (25.2)	38 (33.0)
Race and ethnicity, No. (%)		
White	114 (99)	107 (93)
Nonwhite	1 (1)	8 (7)
BMI (kg/m^2), mean (SD), No.	26.5 (4.1), 115	26.4 (4.2), 115
Clinical, No./total No. (%)		
History of atrial fibrillation	48/115 (41.7)	47/115 (40.9)
History of pulmonary hypertension		
None	100/114 (87.7)	94/110 (85.5)
Moderate (PA systolic 31–55 mm Hg)	7/114 (6.1)	10/110 (9.1)
Severe (PA systolic >55 mm Hg)	7/114 (6.1)	6/110 (5.5)
History of heart failure	30/115 (26.1)	28/115 (24.3)
History of renal failure	4/115 (3.5)	3/115 (2.6)
History of diabetes	7/115 (6.1)	9/115 (7.8)
History of stroke	6/115 (5.2)	9/115 (7.8)
History of previous myocardial infarction	4/115 (3.5)	2/115 (1.7)
History of chronic obstructive pulmonary disease	10/115 (8.7)	9/115 (7.8)
Asthma	12/115 (10.4)	9/115 (7.8)
History of peripheral vascular disease	4/115 (3.5)	0/115 (0.0)
NYHA functional class III/IV	64/114 (56.1)	56/110 (50.9)
Euroscore II, mean (SD), No.	1.47 (0.99), 114	1.58 (1.37), 111
Baseline physical function (SF-36v2 score)		
Low (0 to ≤33)	20 (17.4)	23 (20.0)
Medium (>33 to ≤66)	44 (38.3)	46 (40.0)
High (>66)	51 (44.3)	46 (40.0)

BMI indicates body mass index; NYHA, New York Heart Association; PA, pulmonary artery; and SF-36v2, 36-item Short Form Health Survey version 2.0.

Time (minutes per day) spent in MVPA, light physical activity, and sedentary time were classified using cut points validated for adult populations in combination with arm-angle inclination to distinguish light physical activity from sedentary time.[14,15,17] Total activity counts were extracted from the accelerometry files and summed across waking hours, serving as a proxy for the overall volume of physical activity accumulated in a day.[18] All variables were averaged over the number of valid (ie, 24-hour) measurement days. Accelerometry data sets were considered valid if participants provided ≥4 complete (24-hour) days of measurement.[19,20]

Statistical Analysis

This was an exploratory analysis with prespecified outcomes as defined in the statistical analysis plan (Supplemental Material). All outcomes were analyzed using linear mixed-effects models with adjustment for minimization factors (tricuspid regurgitation, atrial fibrillation, and baseline SF-36v2 score) in the main trial. Models accounted for intraparticipant correlations using random intercepts. Time-based variables (MVPA, light physical activity, sedentary time, and sleep duration) and total activity counts were analyzed as continuous variables. As sleep efficiency follows a beta distribution, beta regression models were used to analyze sleep efficiency. There was no correction for multiple comparisons as these analyses were exploratory.

To assess the impact of missing accelerometry data caused by attrition over time, sensitivity analyses were conducted using multiple imputation, assuming the data were missing at random. Fifteen data sets were multiply imputed for each outcome variable, taking the longitudinal nature of the data into account, using 2-level predictive mean matching to ensure all imputed values were positive. All the variables included in the mixed-effects models were included in the imputation except baseline SF-36v2 scores caused by issues with multicollinearity. The parameters were estimated separately for each imputed data set and combined using Rubin's rules. All analyses were conducted using R version 4.3.1. Multiple imputation was conducted using the mice package (version 3.16.0).

RESULTS

Sample Characteristics

A total of 230 of 330 randomized patients (115 in each arm) took part in the accelerometer substudy and provided valid accelerometer data for at least one time point. Data were unavailable for all 330 patients for one of the following reasons: the substudy was introduced 1 year after the main trial had begun recruiting (n=60); in-house urgent patients were excluded because of the

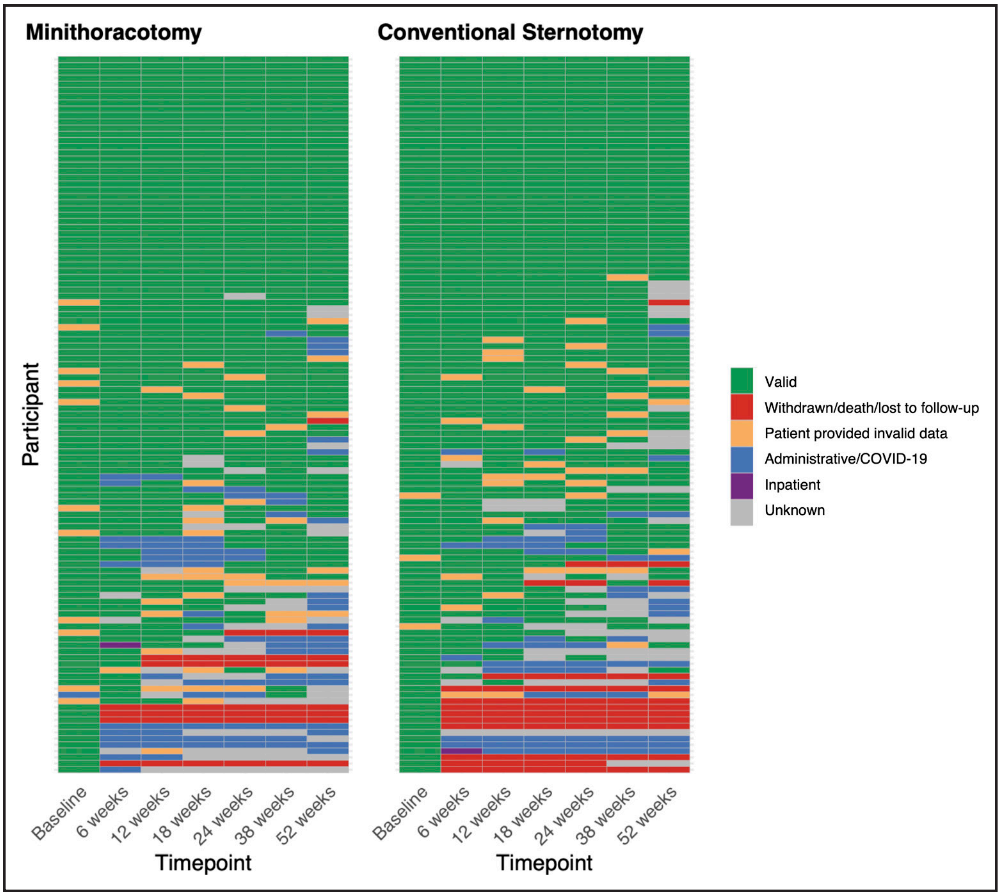

Figure 1. Heatmap visualizing valid and missing accelerometer files for each participant at each time point.

Table 2. Summary and Analysis of Total Activity Counts

	Mini				Sternotomy				Mini - sternotomy	
Time Point	No.	Mean (SD)	Change from baseline difference (95% CI)	*P* value	No.	Mean (SD)	Change from baseline difference (95% CI)	*P* value	Change from baseline difference (95% CI)	*P* value
Baseline	97	304 170 (93 662)	Reference		106	311 899 (97 542)	Reference		Reference	
6 weeks	86	281 226 (91 465)	−32 744 (−45 138 to −20349)	<0.001	81	247 491 (72 899)	−62 436 (−76 548 to −48 323)	<0.001	26 744 (9085 to 44 403)	0.003
12 weeks	80	293 176 (83 896)	−12 365 (−25 116 to 386)	0.058	81	286 801 (89 613)	−28 936 (−43 070 to −14 801)	<0.001	13 590 (−4351 to 31 532)	0.14
18 weeks	67	313 006 (87 151)	11 742 (−1734 to 25 218)	0.088	68	304 296 (93 966)	−17 095 (−32 000 to −2189)	0.025	26 060 (6971 to 45 149)	0.008
24 weeks	67	316 421 (97 344)	12 454 (−926 to 25 834)	0.069	70	316 149 (98 042)	−989 (−15 725 to 13 747)	0.90	10 727 (−8188 to 29 642)	0.27
38 weeks	73	317 191 (97 912)	10 497 (−2585 to 23 579)	0.12	67	304 253 (96 743)	−1022 (−15 983 to 13 938)	0.89	8555 (−10 261 to 27 371)	0.37
52 weeks	61	317 955 (89 928)	11 558 (−2249 to 25 366)	0.10	64	288 635 (89 517)	−8485 (−23 702 to 6733)	0.28	16 967 (−2612 to 36 546)	0.090

lack of opportunity to gather baseline data (n=33); and a small number of patients refused to take part (n=7). A description of the participants who took part in the accelerometry substudy are shown in Table 1.

The number of valid accelerometry files available at each time point ranged from 133 to 215. Sixty-three of the 230 accelerometry participants (27%) provided data at all 7 time points. Twenty accelerometry (8.7%) participants withdrew (n=14), died (n=5), or were lost to follow-up (n=1) during the study (Figure 1). Other reasons for missing data at specific time points during the trial included halting accelerometry data collection because of COVID-19 (affecting n=50 participants at ≥1 time points), administrative reasons (eg, no accelerometer was available to administer, affecting n=17 participants at ≥1 time points), patients being under inpatient care at 6 weeks (n=2 participants), or unknown reasons (n=60 participants at ≥1 time points). The remainder of missing data were missing because of invalidity (affecting n=47 participants at ≥1 time points) for 1 of 3 reasons: the accelerometer battery died during the recording period, insufficient wear, or implausibility (eg, showing MVPA for 24 hours per day).

Impact of Surgical Approach on Accelerometry Outcomes

Total activity counts per day decreased from baseline to 6 weeks in both arms (Table 2; Figure 2); counts remained significantly lower than baseline through 18 weeks in the

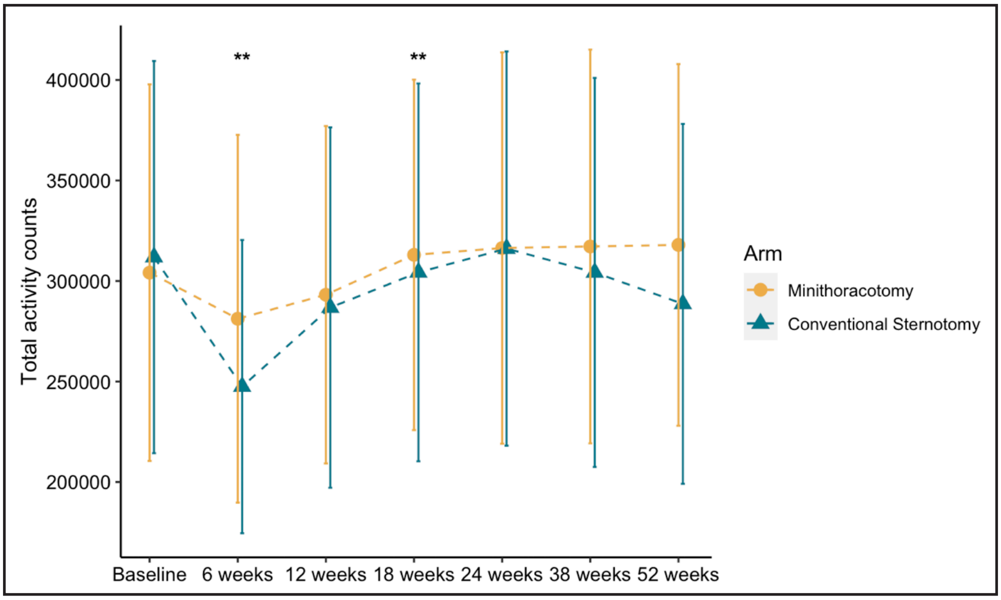

Figure 2. Mean total activity counts per day over time.
Asterisks indicate significant differences between arms based on mixed-effects models (***P*<0.01). Error bars represent SDs.

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Table 3. Summary and Analysis of Daily Time Spent in MVPA, Light Physical Activity, and Sedentary Time

Time Point	Mini				Sternotomy				Mini - sternotomy	
	No.	Mean (SD)* or median (IQR)†	Change from baseline difference (95% CI)	*P* value	No.	Mean (SD)* or median (IQR)†	Change from baseline difference (95% CI)	*P* value	Change from baseline difference (95% CI)	*P* value
Time spent in MVPA (minutes per day)†										
Baseline	103	58 (32 to 99)	Reference		112	59 (31 to 100)	Reference		Reference	
6 weeks	95	42 (25 to 75)	−15 (−21 to −9.1)	<0.001	89	34 (15 to 58)	−31 (−38 to −24)	<0.001	15 (5.7 to 24)	0.001
12 weeks	85	56 (24 to 90)	−7.9 (−14 to −1.5)	0.016	84	46 (27 to 80)	−16 (−23 to −8.9)	<0.001	7.5 (−1.7 to 17)	0.11
18 weeks	73	63 (37 to 93)	1.1 (−5.6 to 7.8)	0.75	78	49 (28 to 94)	−9.6 (−17 to −2.1)	0.012	9.9 (0.31 to 20)	0.043
24 weeks	77	70 (35 to 106)	2.6 (−4.0 to 9.2)	0.44	76	53 (31 to 97)	−6.1 (−14 to 1.4)	0.11	7.9 (−1.6 to 17)	0.10
38 weeks	81	69 (37 to 110)	1.7 (−4.8 to 8.3)	0.60	74	45 (27 to 99)	−3.0 (−11 to 4.6)	0.44	3.9 (−5.6 to 13)	0.42
52 weeks	67	64 (39 to 102)	0.06 (−6.8 to 7.0)	0.99	66	43 (21 to 96)	−5.9 (−14 to 1.9)	0.14	5.1 (−4.9 to 15)	0.32
Time spent in light physical activity (minutes per day)*										
Baseline	98	281 (100)	Reference		109	292 (87)	Reference		Reference	
6 weeks	88	275 (94)	−9.3 (−24 to 5.2)	0.21	82	266 (98)	−25 (−42 to −7.6)	0.005	12 (−8.8 to 33)	0.26
12 weeks	80	287 (100)	3.2 (−12 to 18)	0.67	81	293 (90)	−3.5 (−21 to 14)	0.69	3.2 (−18 to 25)	0.77
18 weeks	68	306 (111)	17 (1.6 to 33)	0.032	70	274 (85)	−20 (−38 to −1.7)	0.033	34 (12 to 57)	0.003
24 weeks	70	301 (114)	19 (3.1 to 34)	0.019	71	293 (91)	−0.41 (−19 to 18)	0.96	15 (−6.9 to 38)	0.18
38 weeks	73	286 (103)	5.7 (−9.6 to 21)	0.46	68	284 (92)	−4.8 (−23 to 14)	0.61	7.0 (−15 to 29)	0.54
52 weeks	61	308 (108)	15 (−0.99 to 31)	0.066	65	279 (92)	−6.8 (−26 to 12)	0.48	19 (−4.3 to 42)	0.11
Time spent sedentary (minutes per day)*										
Baseline	98	641 (110)	Reference		109	619 (111)	Reference		Reference	
6 weeks	88	645 (106)	15 (−2.5 to 33)	0.093	82	667 (104)	47 (26 to 68)	<0.001	−23 (−48 to 2.2)	0.074
12 weeks	80	625 (98)	−9.5 (−28 to 8.9)	0.31	81	639 (103)	26 (4.7 to 47)	0.017	−27 (−52 to −1.1)	0.041
18 weeks	68	614 (100)	−20 (−39 to −0.47)	0.045	70	640 (103)	25 (3.1 to 48)	0.026	−37 (−64 to −9.7)	0.008
24 weeks	70	616 (111)	−17 (−36 to 2.5)	0.090	71	637 (114)	21 (−1.5 to 43)	0.069	−29 (−56 to −1.7)	0.037
38 weeks	73	642 (107)	3.0 (−16 to 22)	0.76	68	653 (103)	29 (6.1 to 51)	0.013	−17 (−44 to 10)	0.23
52 weeks	61	613 (102)	−10 (−30 to 10)	0.33	65	637 (113)	14 (−8.9 to 37)	0.23	−15 (−43 to 13)	0.29

IQR indicates interquartile range; and MVPA, moderate-to-vigorous physical activity.
*Mean (SD).
†Median (IQR).

sternotomy arm only (Table 2). The difference in mean change in activity counts differed between arms at 6 weeks and 18 weeks, favoring Mini by 26 744 (95% CI, 9085–44 403) and by 26 060 (95% CI, 6971–45 149) at the respective time points (Table 2; Figure 2).

Following a similar pattern as total activity counts, time spent in MVPA remained significantly lower than baseline through 18 weeks in the sternotomy arm, whereas in the Mini arm, it remained lower through 12 weeks only (Table 3). The difference in mean change in MVPA minutes favored the Mini arm by 15 minutes at 6 weeks and by 9 minutes at 18 weeks (Table 3; Figure 3). In terms of percent change, time spent in MVPA decreased from baseline to 6 weeks by a median of 46.0% in the sternotomy arm versus 19.7% in the Mini arm; by 12 weeks, percent change from baseline was similar in both arms, with sternotomy and Mini being 15.4% and 15.2% below baseline values, respectively.

That total activity counts and MVPA did not exceed baseline in either arm in the 52 weeks after surgery was unexpected, particularly given that the majority of participants were symptomatic at baseline. We therefore conducted exploratory post hoc subgroup analyses to determine whether the trajectory in physical activity differed by baseline physical activity level (> or ≤ median values), based on literature indicating an inverse relationship between preoperative physical activity levels and activity levels up to 1 year after cardiac surgery.[21] For MVPA, the interaction between time, arm, and baseline activity was significant at multiple time points (6, 12, 18, and 52 weeks), indicating MVPA increased over time among those in the Mini arm with below-median preoperative MVPA; the MVPA of more active counterparts did not exceed baseline values at any time point (Figure 4). The pattern for total activity counts was visually similar, but the interactions were nonsignificant (P>0.05) at all time points (Figure 4).

In general, compared with baseline, time spent in light physical activity increased from 12 weeks onward in the Mini arm, whereas it decreased in the sternotomy arm,

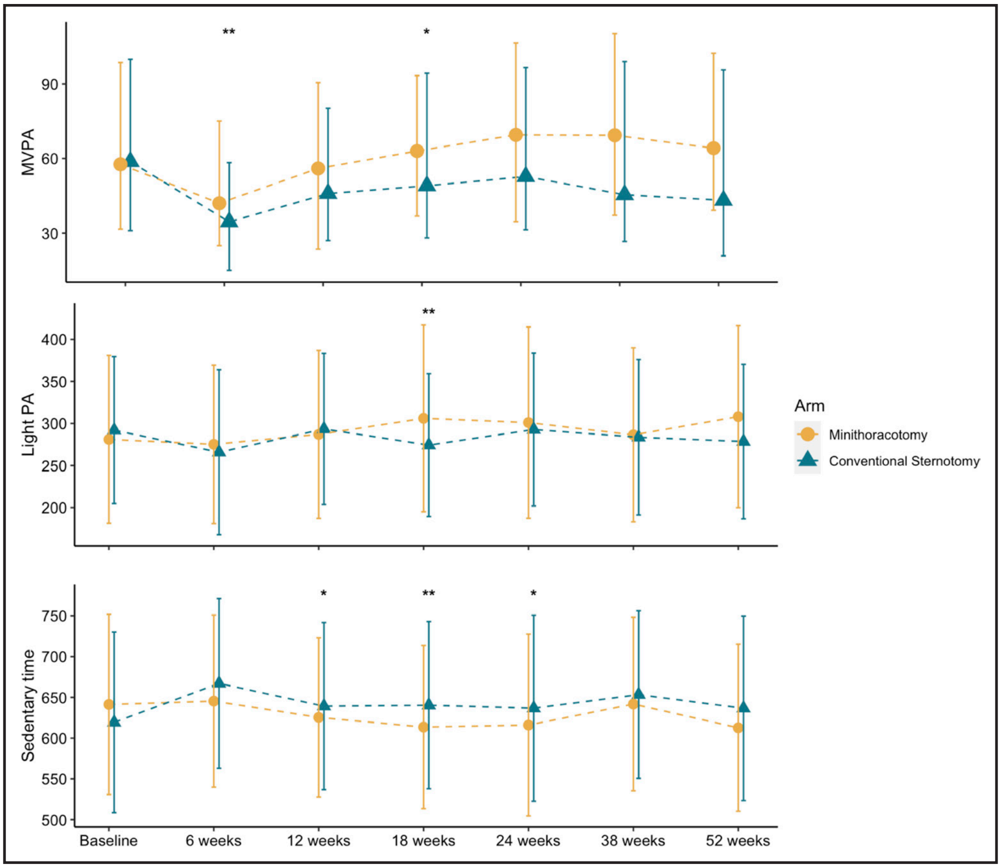

Figure 3. Median time spent in MVPA and mean time in light physical activity and sedentary time per day over time.
Asterisks indicate significant differences between arms based on mixed-effects models (*P<0.05; **P<0.01). Error bars represent interquartile ranges (MVPA) or SDs (light PA and sedentary time). MVPA indicates moderate-to-vigorous physical activity; and PA, pulmonary artery.

with significant between-arm differences at 18 weeks (Table 3; Figure 3). Sedentary time increased after surgery in the sternotomy arm, but not in the Mini arm, with significant differences favoring Mini at 12, 18, and 24 weeks (Table 3; Figure 3). There were no differences in sleep duration between arms at any of the time points (Table 4; Figure 5). Sleep efficiency was significantly lower than baseline at all time points among those in the sternotomy arm (Table 4). The difference in mean change in sleep efficiency favored the Mini arm, with a significant difference between arms at 12 weeks (Table 4). Sensitivity analyses with the imputation of missing data showed broadly similar results, particularly for MVPA and sleep efficiency, favoring the Mini arm at 6 and 12 weeks, respectively (Table S1). There were no differences between arms in total activity counts, light physical activity, or sedentary time in the sensitivity analyses.

DISCUSSION

Very few studies have used accelerometry in the context of cardiac surgery.[22,23] To our knowledge, this is the first study that has compared accelerometer-measured physical activity according to cardiac surgical interventions in a randomized controlled trial setting. The data confirm that accelerometry can be used to chart recovery of physical function after surgery. Of particular note is that no differences in self-reported physical functioning from baseline to 12 weeks were found in the main findings of the trial,[5] but differences in accelerometer-measured physical activity were found here. This underlines the value of using objective measures such as accelerometry to capture changes in physical activity and functioning over time.

Although there was no improvement in physical activity in either arm compared with baseline, the findings show that the minimally invasive approach leads to an earlier recovery of total physical activity and MVPA. This is consistent with the findings of a nonrandomized study that similarly showed greater reductions in postoperative accelerometer-measured physical activity among sternotomy patients compared with patients who underwent minimally invasive cardiac surgery.[23] Exploratory subgroup analyses indicated there were improvements in MVPA over time among those in the Mini arm who were least active before surgery. Although this analysis was post hoc and should be interpreted with caution, it is consistent with similar findings reported previously suggesting that the impact of surgery on physical activity may be more pronounced among those whose preoperative

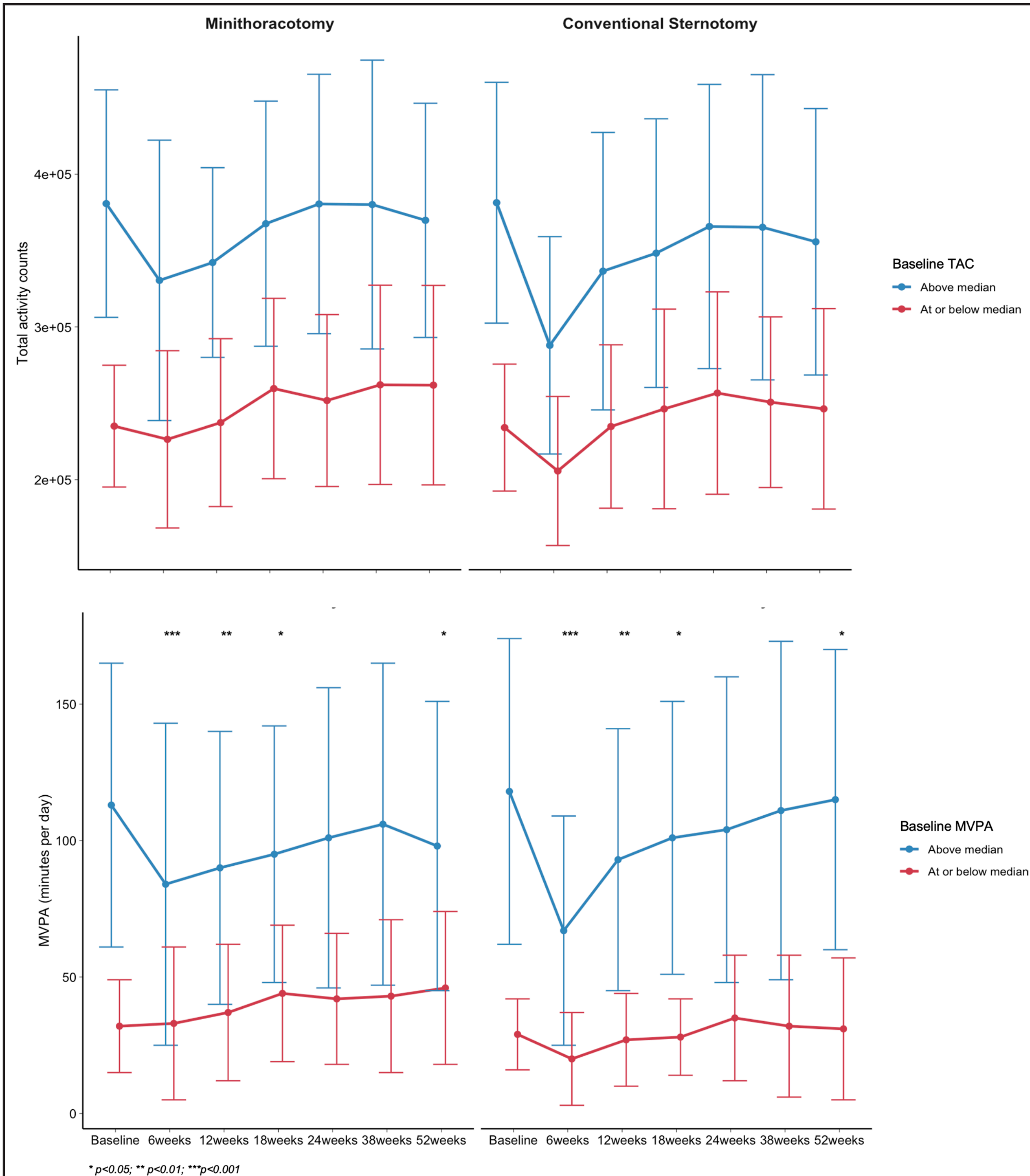

Figure 4. Exploratory subgroup analysis of total activity counts and mean time spent in MVPA over time stratified by arm and baseline activity counts or MVPA.

Asterisks indicate significant arm by time by subgroup interactions based on mixed-effects models (**P*<0.05; ***P*<0.01). MVPA indicates moderate-to-vigorous physical activity.

physical activity was low, perhaps because of relief of their symptoms.[21]

The findings also indicated differences between arms in light physical activity and sedentary time, with patients in the Mini arm tending to have more light physical activity and less sedentary time up to 24 weeks after surgery compared with patients in the sternotomy arm. These findings are not necessarily distinctive from each other, as the nature of time-use data means that time spent in one behavior

Table 4. Summary and Analysis of Sleep Duration and Sleep Efficiency

	Mini				Sternotomy				Mini - sternotomy	
Variable/ Time Point	No.	Mean (SD)* or median (IQR)†	Change from baseline difference (95% CI)	*P* Value	No.	Mean (SD)* or median (IQR)†	Change from baseline difference (95% CI)	*P* Value	Change from baseline difference (95% CI)	*P* Value
Sleep duration (minutes per day)*										
Baseline	98	444 (62)	Reference		109	454 (58)	Reference		Reference	
6 weeks	88	453 (72)	4.3 (−7.5 to 16)	0.48	82	460 (77)	6.5 (−8.4 to 21)	0.39	−6.5 (−24 to 11)	0.46
12 weeks	80	457 (65)	10 (−1.8 to 23)	0.10	81	447 (83)	−6.7 (−22 to 8.3)	0.38	13 (−4.8 to 30)	0.15
18 weeks	68	444 (68)	−3.9 (−17 to 8.9)	0.55	70	451 (57)	−1.8 (−18 to 14)	0.82	−6.5 (−25 to 12)	0.49
24 weeks	70	441 (67)	−7.1 (−20 to 5.6)	0.27	71	439 (67)	−14 (−30 to 1.2)	0.072	3.0 (−15 to 21)	0.75
38 weeks	73	440 (69)	−6.2 (−19 to 6.4)	0.34	68	436 (54)	−20 (−36 to −3.7)	0.016	8.9 (−9.5 to 27)	0.34
52 weeks	61	441 (64)	−8.4 (−22 to 4.9)	0.22	65	467 (91)	4.7 (−11 to 21)	0.57	−18 (−37 to 1.4)	0.070
Sleep efficiency (proportion of sleep period spent asleep)†										
Baseline	98	0.62 (0.53 to 0.74)	Reference		109	0.68 (0.51 to 0.76)	Reference		Reference	
6 weeks	88	0.63 (0.51 to 0.76)	−0.06 (−0.22 to 0.10)	0.47	82	0.60 (0.50 to 0.73)	−0.20 (−0.36 to −0.05)	0.012	0.13 (−0.06 to 0.32)	0.19
12 weeks	80	0.67 (0.54 to 0.75)	0.02 (−0.14 to 0.19)	0.77	81	0.59 (0.45 to 0.72)	−0.27 (−0.43 to −0.11)	<0.001	0.27 (0.07 to 0.47)	0.007
18 weeks	68	0.62 (0.51 to 0.73)	−0.12 (−0.29 to 0.05)	0.18	70	0.58 (0.46 to 0.76)	−0.23 (−0.40 to −0.07)	0.006	0.09 (−0.12 to 0.30)	0.38
24 weeks	70	0.63 (0.53 to 0.75)	−0.04 (−0.21 to 0.13)	0.67	71	0.62 (0.50 to 0.70)	−0.17 (−0.34 to −0.01)	0.042	0.11 (−0.10 to 0.32)	0.29
38 weeks	73	0.64 (0.54 to 0.74)	−0.03 (−0.19 to 0.14)	0.76	68	0.62 (0.45 to 0.70)	−0.25 (−0.42 to −0.08)	0.004	0.20 (−0.01 to 0.41)	0.065
52 weeks	61	0.62 (0.45 to 0.74)	−0.17 (−0.35 to 0.00)	0.055	65	0.64 (0.51 to 0.73)	−0.24 (−0.41 to −0.07)	0.006	0.03 (−0.19 to 0.25)	0.79

IQR indicates interquartile range.
*Mean (SD).
†Median (IQR).

(eg, physical activity) directly displaces time spent in another behavior (eg, sedentary time).[24,25] Overall, the data indicate that those in the Mini arm were able to engage in physical activity earlier after surgery than those in the sternotomy arm, perhaps through returning to work and social hobbies as well as recovery activities such as physiotherapy exercises and other postdischarge mobility activities.

No differences in sleep duration were observed between trial arms, but sleep efficiency was higher at 12 weeks in the Mini arm versus the sternotomy arm. This suggests that the Mini procedure may have a smaller impact on sleep quality than sternotomy.

These findings have potential implications on patients and hospitals. When faced with decisions about surgery, patients are often worried about recovery time. A Mini approach appears to have a smaller and shorter impact on patients' physical activity and functioning. This aligns with previously reported findings from the UK Mini Mitral trial and elsewhere, which showed that patients who underwent a Mini approach had shorter hospital stays (Figure 6), were more likely to be discharged from the hospital early, and had more days alive and out of hospital at both 30 and 90 days after surgery compared with those who underwent sternotomy.[5,26] The reduced impact of Mini on physical functioning may influence decision-making among both young and elderly patients, and also has important implications for health systems, in which the number of available beds are limited.

These findings also have implications for future randomized controlled trials comparing 2 interventions or indeed trials comparing interventions versus medical therapy in which outcomes relating to physical function are important. Data collected from wearable devices analyzed by blinded researchers are less prone to bias and measurement error compared with self-reported measures. We believe this is the first large multicenter randomized controlled trial showing the benefits of analyzing data captured from wearable data within the context of cardiac surgery and would encourage more researchers to consider including accelerometry when planning future trials.

This study has several limitations. First, this study was powered to detect changes in the primary outcome, SF-36v2 physical functioning score (reported in Akowuah et al[5]), and thus may not have sufficient statistical power to

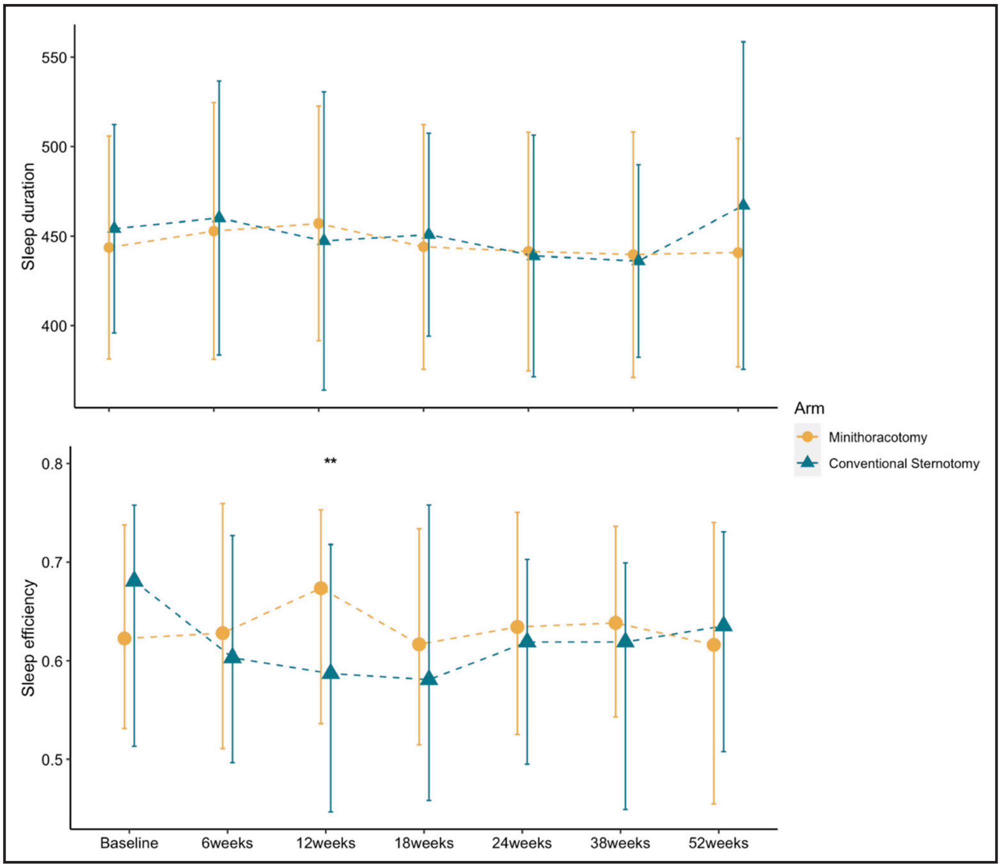

Figure 5. Mean sleep duration and median sleep efficiency per day over time.
Asterisks indicate significant differences between arms based on mixed-effects models (**$P<0.01$). Error bars represent SDs (sleep duration) or interquartile ranges (sleep efficiency).

detect differences in physical activity variables between groups. Second, the longitudinal data analysis approach used here, which involves the interpretation of many *P* values, raises the possibility of an increased type 1 error risk. Third, as with most surgical trials, the trial itself was not blinded (ie, patients and surgeons were aware of allocation),

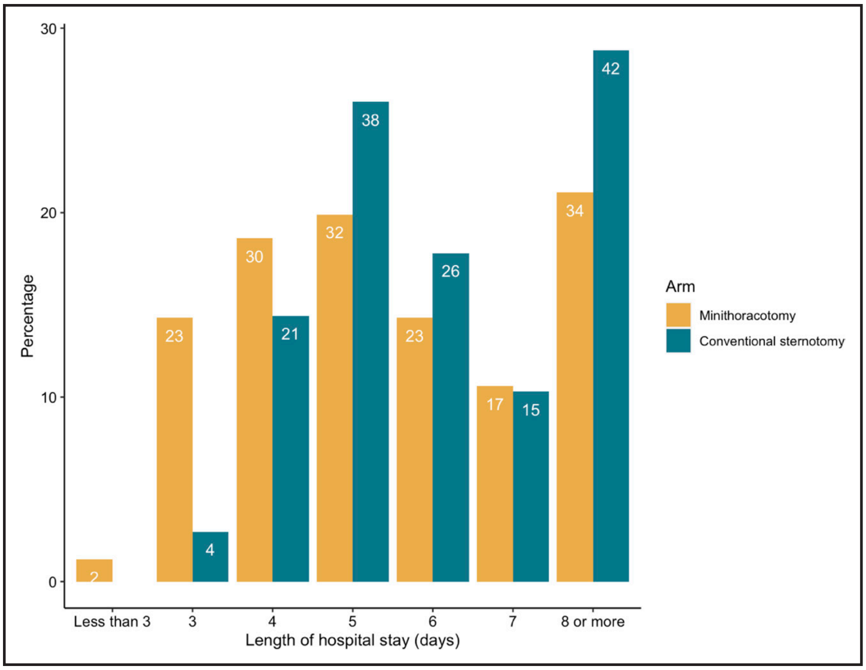

Figure 6. Length of hospital stay after index surgery (data from Akowuah et al[5]).
Data are shown as percentages, with original counts shown on the bars.

which poses a risk of bias. However, the accelerometry data processing and analysis were conducted by blinded researchers. Additionally, reactivity in free-living accelerometry studies has been suggested to be low (especially for MVPA),[27] indicating the likelihood that patients altered their behavior in response to the presence of the accelerometer or knowledge of their allocation is probably low. Fourth, the cut points applied to the accelerometry data are based on healthy adult populations and may not be suitable for a clinical population, although the similarities between findings for total activity count (which do not require application of cut points), and time spent in MVPA suggests these cut points may have been suitable. Fifth, this study used wrist-worn accelerometers, which have limitations in the measurement of sedentary time.[16] Last, this study recruited the majority of patients during the COVID-19 pandemic, which may have affected physical activity, although as this was a randomized study, such impacts would have likely affected both arms to a comparable extent.

Conclusions

This analysis of data collected within the UK Mini Mitral trial has demonstrated that accelerometry provides additional and complementary data on the recovery of physical activity and sleep after surgery. The data suggest a greater early drop in physical activity in patients undergoing a sternotomy approach. It also suggests an earlier recovery of physical activity and better sleep efficiency among patients undergoing a Mini approach for mitral valve repair surgery.

ARTICLE INFORMATION

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Supplemental Material

Statistical Analysis Plan
Table S1

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